

Session 1: Univariate and Joint Tests Using `lincom` and `testparm`

Intuition for Wald vs. LR Tests

January 2026

1 Purpose and Scope

These notes organize a complete learning pipeline around a small set of core Stata commands (`reg`, `test`, `lincom`, `testparm`, and `lrtest`)—from data preparation, to regression, to interpretation, to hypothesis testing, and finally to producing clean outputs. The guiding principle throughout is not only to show *how* to run each command, but also to explain *why* the step is conceptually necessary.

2 Using Chinese Firm Data for Demos

A minimal template (edit variable names as needed)

```
*****
* Using a cleaned file with:
* go = gross output (current price)
* M = intermediate inputs
* L = labor (avg employment)
* K = capital (avg net fixed assets)
*****

use "1998_clean.dta", clear

gen lnGO = ln(go) if go>0
gen lnM = ln(M) if M>0
gen lnL = ln(L) if L>0
gen lnK = ln(K) if K>0
drop if missing(lnGO, lnM, lnL, lnK)

reg lnGO lnL lnK lnM

test lnL = 0
test lnL lnK
lincom lnL + lnK

* Returns to scale (CRS) in GO setting with intermediates
test lnL + lnK + lnM = 1
lincom lnL + lnK + lnM - 1
```

Results

```

. reg lnGO lnL lnK lnM

Source | SS df MS Number of obs = 156,258
-----+----- F(3, 156254) > 99999.00
Model | 355655.72 3 118551.907 Prob > F = 0.0000
Residual | 26472.993 156,254 .169422818 R-squared = 0.9307
-----+----- Adj R-squared = 0.9307
Total | 382128.713 156,257 2.4455142 Root MSE = .41161

-----
lnGO | Coefficient Std. err. t P>|t| [95% conf. interval]
-----+-----
lnL | .0675274 .0012291 54.94 0.000 .0651184 .0699365
lnK | .0185919 .0008214 22.63 0.000 .0169819 .0202019
lnM | .8872571 .0008332 1064.85 0.000 .885624 .8888902
_cons | .832835 .0061638 135.12 0.000 .8207541 .844916
-----

.
. test lnL = 0

( 1) lnL = 0

F( 1,156254) = 3018.37
Prob > F = 0.0000

. test lnL lnK

( 1) lnL = 0
( 2) lnK = 0

F( 2,156254) = 3338.52
Prob > F = 0.0000

. lincom lnL + lnK

( 1) lnL + lnK = 0

-----
lnGO | Coefficient Std. err. t P>|t| [95% conf. interval]
-----+-----
(1) | .0861193 .0010608 81.18 0.000 .0840402 .0881984
-----

.
. * Returns to scale (CRS) in GO setting with intermediates
. test lnL + lnK + lnM = 1

( 1) lnL + lnK + lnM = 1

F( 1,156254) = 956.12
Prob > F = 0.0000

```

```
. lincom lnL + lnK + lnM - 1
```

```
( 1) lnL + lnK + lnM = 1
```

```
-----
      lnGO | Coefficient Std. err. t P>|t| [95% conf. interval]
-----+-----
      (1) |  -.0266237   .000861 -30.92 0.000  -.0283112  -.0249361
-----

.
end of do-file
```

This estimates a log–log “production” relationship:

$$\ln(GO_i) = \alpha + \beta_L \ln(L_i) + \beta_K \ln(K_i) + \beta_M \ln(M_i) + u_i.$$

In a log–log model, coefficients are elasticities (*ceteris paribus*):

- **Labor elasticity** $\hat{\beta}_L = 0.0675$: holding K and M fixed, a 1% increase in labor is associated with about a 0.0675% increase in gross output.
- **Capital elasticity** $\hat{\beta}_K = 0.0186$: holding L and M fixed, a 1% increase in capital is associated with about a 0.0186% increase in gross output.
- **Intermediate input elasticity** $\hat{\beta}_M = 0.8873$: holding L and K fixed, a 1% increase in intermediates is associated with about a 0.887% increase in gross output.

Key economic takeaway. Output (GO) is overwhelmingly tied to intermediate inputs in this specification; that is why β_M is close to 1.

Model fit

$R^2 = 0.9307$ is very high—again consistent with GO being tightly linked to intermediates in accounting-style data.

About p-values

With $N \approx 156,000$, standard errors are tiny, so almost everything becomes “statistically significant.” Focus on magnitudes.

Interpreting tests: test $\ln L = 0$ and test $\ln L \ln K$

test $\ln L = 0$ Null:

$$H_0 : \beta_L = 0.$$

We obtain $F = 3018.37$ with $p = 0.0000$, so reject H_0 . The labor elasticity is statistically different from zero.

test $\ln L \ln K$ Null:

$$H_0 : \beta_L = 0 \quad \text{and} \quad \beta_K = 0 \quad (\text{jointly}).$$

We obtain $F = 3338.52$ with $p = 0.0000$, so reject H_0 . Labor and capital are jointly significant.

```
lincom lnL + lnK
```

This computes $\hat{\beta}_L + \hat{\beta}_K$ and tests the default null

$$H_0 : \beta_L + \beta_K = 0.$$

Result:

$$\hat{\beta}_L + \hat{\beta}_K = 0.0861.$$

Interpretation: the combined elasticity of primary inputs (L and K) is about 0.086 in this *GO*-with-intermediates setup (not a “returns to scale” number by itself, because M is still in the model).

Returns-to-scale test: test lnL + lnK + lnM = 1

This is the meaningful CRS restriction in a Cobb–Douglas interpretation with all three inputs:

$$H_0 : \beta_L + \beta_K + \beta_M = 1.$$

The estimated sum is

$$0.0675 + 0.0186 + 0.8873 \approx 0.9734,$$

i.e., below 1.

The test output gives $F = 956.12$ with $p = 0.0000$, so you reject CRS.

```
lincom lnL + lnK + lnM - 1
```

This reports the estimated deviation from 1:

$$(\hat{\beta}_L + \hat{\beta}_K + \hat{\beta}_M) - 1 = -0.02662,$$

with a tight confidence interval around -0.027 .

Interpretation: statistically, your estimated returns-to-scale parameter is about 0.973, which is slightly decreasing returns to scale in this specification.

Important nuance. With a dataset this large, “statistically different from 1” is easy to obtain even for small deviations. The economic size here is a 2.7% deviation from CRS—small, though precisely estimated.

Summary

Using 1998 Chinese industrial firms, a log–log *GO* regression shows intermediates have elasticity ≈ 0.887 , labor ≈ 0.068 , capital ≈ 0.019 , and the CRS restriction $\beta_L + \beta_K + \beta_M = 1$ is rejected in favor of a precisely estimated sum of about 0.973 (slightly decreasing returns to scale).

3 Core Example: Industry Effects in nlsw88

We use Stata’s built-in dataset `nlsw88` to illustrate how a categorical variable (industry) enters a regression and how to test whether industry effects are jointly significant.

3.1 From a categorical variable to dummies: `tab industry, gen(d)`

In many applications, `industry` is a categorical variable with multiple levels (here, 12). A regression needs numeric regressors, so we represent industry membership by a set of 0/1 indicator variables (dummies). The command `tab industry, gen(d)` creates `d1–d12`, where each observation has exactly one dummy equal to 1 and the rest equal to 0.

When the regression includes an intercept `_cons`, we cannot include all 12 dummies simultaneously because they sum to one:

$$d1 + d2 + \dots + d12 = 1,$$

which is perfectly collinear with the intercept. This is the classic “dummy variable trap.” The standard fix is to omit exactly one category, which becomes the reference (base) group. In the manual-dummy approach, this is done explicitly via `drop d1`, making industry 1 the base category. All remaining industry coefficients are then interpreted as differences relative to industry 1.

Code

```
sysuse "nls88.dta", clear
tab industry, gen(d)
drop d1 // Set industry 1 as the base group
reg wage age tenure married collgrad d*
```

3.2 What model is being estimated? (One model, not one model per person)

The command

```
reg wage age tenure married collgrad d*
```

estimates a **single** OLS model (one set of coefficients) on the whole sample:

$$wage_i = \alpha + \beta_1 age_i + \beta_2 tenure_i + \beta_3 married_i + \beta_4 collgrad_i + \sum_{j=2}^{12} \gamma_j d_{ji} + u_i.$$

Each row in the dataset is one observation (one person), but OLS does *not* fit a separate regression per person. Instead, it chooses one coefficient vector to minimize the sum of squared residuals across all observations. Predicted wages differ across individuals because their covariates differ:

$$\widehat{wage}_i = \hat{\alpha} + \hat{\beta}_1 age_i + \dots + \hat{\gamma}_j d_{ji},$$

not because the model changes from person to person.

3.3 Why include controls (age, tenure, marital status, education)?

The goal of adding controls is to avoid mistaking compositional differences across industries for genuine industry effects. Industries differ systematically in worker characteristics (education, experience, etc.), and these characteristics directly affect wages. By including variables such as age, tenure, marital status, and college graduation, the industry coefficients are interpreted closer to a *ceteris paribus* comparison: differences in wages across industries among otherwise similar workers.

4 Testing: What do test and lincom do?

4.1 Joint significance in a production-function example: test lnL lnK

In a production-function regression such as

$$\ln Y = a + b_L \ln L + b_K \ln K + u,$$

the command `test lnL lnK` tests the joint null

$$H_0 : b_L = 0 \text{ and } b_K = 0,$$

i.e., whether the regressors (excluding the intercept) are jointly significant.

4.2 Linear combinations: lincom lnL + lnK and CRS

The command `lincom lnL + lnK` computes and tests a linear combination of coefficients. By default it tests

$$H_0 : b_L + b_K = 0,$$

reporting the estimate, its standard error (computed using the full covariance matrix), a t-statistic, and a confidence interval. In many economics applications, a more meaningful restriction is constant returns to scale in Cobb–Douglas:

$$H_0 : b_L + b_K = 1.$$

This can be tested directly as:

```
test lnL + lnK = 1
* or equivalently
lincom lnL + lnK - 1
```

5 Are industry effects significant? Two equivalent approaches

5.1 Approach A: Manual dummies + test

After setting a base group (e.g., `drop d1`), a standard question is whether industry effects can be removed from the model. This is a joint test of

$$H_0 : \gamma_2 = \gamma_3 = \dots = \gamma_{12} = 0.$$

In Stata:

```
sysuse "nls88.dta", clear
tab industry, gen(d)
drop d1
reg wage age tenure married collgrad d*
test d2=d3=d4=d5=d6=d7=d8=d9=d10=d11=d12=0
```

A significant F-statistic (e.g., $F(11, 2201) = 12.25$ with $p < 0.001$) implies that industry dummies, as a group, provide significant explanatory power even after controlling for worker characteristics. The numerator degrees of freedom (11) correspond to the 11 included industry dummies (excluding the base group), and the denominator degrees of freedom correspond to the residual degrees of freedom.

5.2 Approach B: Factor-variable syntax + testparm (recommended)

A cleaner and less error-prone approach uses Stata's factor-variable notation. The term `i.industry` instructs Stata to generate the appropriate dummy variables automatically while handling the base category correctly. The notation `ib2.industry` sets industry 2 as the reference group (base). The command `testparm i.industry` then performs the joint test that all non-base industry indicators are zero:

$$H_0 : \gamma_j = 0 \quad \forall j \neq 2.$$

```
sysuse "nlsw88.dta", clear
local cx "age_tenure_married_collgrad"
reg wage 'cx' ib2.industry
testparm i.industry
```

Changing the base category affects only the interpretation (differences relative to the chosen reference group), not fitted values, residuals, R^2 , or the conclusion of the joint significance test.

6 Supplement: LR test (lrtest) and why if e(sample) matters

In nested-model comparisons, another common test is the likelihood ratio (LR) test. The idea is to estimate a *large* model and a *restricted* (small) model, then test whether the improvement in fit is statistically significant.

Code (kept close to the original workflow)

```
sysuse "nlsw88.dta", clear
reg wage age tenure married collgrad i.industry
est store m1
reg wage age tenure married collgrad if e(sample)
est store m0 // why?

lrtest m1 m0
return list
global chi2 = r(chi2)
global pval = r(p)
```

Large vs restricted model (nested)

The large model includes industry indicators:

$$wage_i = \alpha + \beta' X_i + \sum_{j \neq \text{base}} \gamma_j \mathbf{1}\{industry_i = j\} + u_i,$$

while the restricted model omits them:

$$wage_i = \alpha + \beta' X_i + u_i,$$

which is equivalent to imposing $\gamma_j = 0$ for all non-base industries.

Why if e(sample)?

When the large model is estimated first, Stata may drop observations due to missing values in any included variable. The indicator `e(sample)` marks the exact set of observations used in that estimation. Running the restricted model with `if e(sample)` ensures that both models are estimated on the **same sample**. This is essential: LR (and any nested-model comparison) is only meaningful when models are fit to identical data.

LR test intuition

The LR statistic compares log-likelihoods:

$$LR = -2(LL_0 - LL_1) = 2(LL_1 - LL_0),$$

where LL_1 is the log-likelihood of the large model and LL_0 is that of the restricted model. The null hypothesis is that the restrictions are valid (i.e., the small model is sufficient). A large increase in log-likelihood after adding industry dummies implies a large LR statistic and leads to rejection, concluding that industry effects are jointly significant.

Wald vs LR: the core intuition

Wald test: estimate the *full model only*; most commonly used in linear models and typically implemented via OLS.

LR test: estimate both the *full model* and the *restricted model*, and test whether the difference between them is statistically significant; mainly used in MLE-based models (e.g., Logit, Probit), especially in nonlinear settings.

In linear OLS regressions, practitioners usually prefer `test/testparm` (Wald/F) for joint restrictions, while `lrtest` is more common in MLE contexts. Under normal-error assumptions, OLS can be framed as MLE, so Stata can report log-likelihood and run LR tests; in many applications, Wald/F and LR lead to consistent conclusions about joint significance.

7 Reporting: Adding the F-test to a regression table (estadd/esttab)

A practical reporting pattern is to attach the joint-test statistic to the estimation object and print it in the regression table. After running `testparm`, the F-statistic and p-value are stored in `r(F)` and `r(p)`. The `estadd` command appends them to the stored estimation result, and `esttab` prints them as scalars at the bottom of the table.

```
return list // Inspect r(F), r(p) returned by testparm
estadd scalar Fvalue = r(F) // Add the F statistic to the estimation object
estadd scalar Fp = r(p) // Add the p-value to the estimation object
est store F_test

esttab F_test, scalar(Fvalue Fp) sfmt(%4.3f) ///
    nogap r2 ar2 nobase drop(*.industry) ///
    addnotes("Fvalue: Joint significance test for industry dummies (F-test)" ///
        "Fp: p-value corresponding to the F-test")
```


8 Univariate and Joint Hypothesis Tests: Input Elasticities, CRS, and Industry Fixed Effects (1998 Chinese Industrial Firms)

This section documents a complete testing workflow for a gross-output specification estimated on the 1998 China Industrial Enterprises data. The objective is twofold: (i) to clarify why we run both univariate (single-parameter) and joint (multi-parameter) tests, and (ii) to show how these tests inform specification choices such as whether to include industry fixed effects (FE). Throughout, inference is reported using standard errors clustered by industry (40 industry clusters in the estimation sample).

8.1 Empirical specification and sample construction

Let GO denote gross output (current price), M intermediate inputs, L labor (average employment), and K capital (average net fixed assets). We estimate the log-log specification

$$\ln(GO_i) = \alpha + \beta_L \ln(L_i) + \beta_K \ln(K_i) + \beta_M \ln(M_i) + u_i, \quad (1)$$

and an augmented version with industry fixed effects:

$$\ln(GO_i) = \alpha + \beta_L \ln(L_i) + \beta_K \ln(K_i) + \beta_M \ln(M_i) + \sum_{j \neq \text{base}} \gamma_j \mathbf{1}\{\text{industry}_i = j\} + u_i. \quad (2)$$

Log variables are constructed only when the underlying levels are strictly positive. Observations with missing $\ln(GO)$, $\ln(M)$, $\ln(L)$, or $\ln(K)$ are dropped, yielding $N = 156,258$ usable observations.

8.2 Why both univariate and joint tests?

Univariate and joint tests answer different questions that are both essential in applied work.

- **Univariate tests** evaluate a single restriction, such as $H_0 : \beta_L = 0$. They are useful for assessing whether a specific regressor has a statistically distinguishable association with the outcome (given the model), or for checking a single theoretical value (e.g., testing $\beta_L = 0.7$).
- **Joint tests** evaluate multiple restrictions simultaneously, such as $H_0 : \beta_L = \beta_K = 0$ or $H_0 : \gamma_1 = \dots = \gamma_J = 0$. These tests answer whether a *set* of regressors is collectively relevant (or whether a structural restriction holds), and they are the natural tool for deciding whether a block of controls (e.g., fixed effects) can be removed without sacrificing explanatory power.

8.3 Empirical results: no industry FE vs industry FE (clustered by industry)

Table 1 reports coefficient estimates from the baseline model (no FE) and the model with industry FE, both estimated with standard errors clustered by industry.

Two patterns are notable. First, the intermediate-input elasticity is close to one in both specifications, indicating that gross output co-moves strongly with intermediate inputs in this accounting-based setting. Second, adding industry FE changes the input elasticities only modestly (labor is essentially unchanged; capital decreases and becomes more precisely estimated), while R^2 increases slightly.

8.4 Joint tests: groups of coefficients and structural restrictions

(i) Joint significance of industry fixed effects. To assess whether industry effects matter *as a block*, we test the null that all non-base industry coefficients are zero:

$$H_0 : \gamma_7 = \gamma_8 = \dots = \gamma_{46} = 0.$$

Table 1: Gross output regression with and without industry FE (SE clustered by industry)

	(0) No industry FE	(1) Industry FE
$\ln(L)$	0.06753 (0.00468)	0.06709 (0.00428)
$\ln(K)$	0.01859 (0.01014)	0.01188 (0.00525)
$\ln(M)$	0.88726 (0.01105)	0.89314 (0.00812)
Constant	0.83284 (0.03130)	0.92591 (0.02214)
N	156,258	156,258
R^2	0.9307	0.9318

Notes. Standard errors (in parentheses) are clustered by industry (40 clusters). Column (1) includes a full set of industry indicator variables (one category omitted as the base group). The overall model F statistic is not always displayed under clustered inference with many indicators; inference on coefficients and explicit hypothesis tests remain valid.

Using Stata’s factor-variable aware joint test,

```
testparm i.industry
```

we obtain

$$F(39, 156215) = 42.12, \quad p < 0.001,$$

which strongly rejects the null. Hence, even after controlling for $\ln(L)$, $\ln(K)$, and $\ln(M)$, average log gross output differs systematically across industries. Practically, this supports including industry FE to absorb persistent cross-industry differences (technology, product mix, accounting conventions, and other industry-level components) that would otherwise be loaded into the error term.

(ii) Constant returns to scale (CRS) in the gross-output setting. With three inputs in a Cobb–Douglas interpretation, the CRS restriction is

$$H_0 : \beta_L + \beta_K + \beta_M = 1.$$

In the industry-FE specification with clustering by industry, the linear restriction test

```
test lnL + lnK + lnM = 1
```

rejects CRS, and the corresponding linear-combination estimate reports

$$(\beta_L + \widehat{\beta_K} + \beta_M) - 1 = -0.0279 \quad (\text{s.e. } 0.0029), \quad F(1, 39) = 94.36, \quad p < 0.001,$$

implying an estimated sum of elasticities around 0.972. The deviation from one is economically small (about 2.8%), but precisely estimated given the large sample.

8.5 Implication for model choice: do we “need” industry FE?

A common practical question is whether industry FE should be included by default. The joint test above rejects the hypothesis that industry FE are collectively zero, so in this application the data strongly support their inclusion. More generally, however, *failing to reject* a joint FE test does not prove the absence of industry heterogeneity; it may reflect low power (e.g., few clusters, noisy data) rather than true equality across industries. A useful rule of thumb is:

- If the FE block is not jointly significant *and* adding FE leaves the coefficients of interest essentially unchanged, a parsimonious specification without FE is often defensible.

- If adding FE materially changes key coefficients, FE may still be valuable as a robustness/control device even if the joint test is weak, because it suggests correlation between unobserved industry-level factors and regressors.

In the 1998 firm data, industry FE are both jointly significant and substantively relevant, while the main elasticities remain stable, indicating that FE primarily absorb level differences across industries rather than overturning the core input-output relationship.

9 Summary: Six Takeaways

1. To include a categorical variable (industry) in a regression, use dummies: `i.industry` or `tab ..., gen(d)`.
2. One category must be the base group to avoid perfect collinearity with the intercept; the choice affects only “relative-to-which-group” interpretation.
3. `reg` estimates **one** coefficient vector for the whole sample, not one model per observation.
4. `test` / `testparm` implement **joint significance tests**: whether a set of coefficients can be set to zero together.
5. `lincom` tests **linear combinations** of coefficients, e.g., returns to scale $b_L + b_K = 1$.
6. Controls help avoid confusing compositional differences with group effects, making comparisons closer to *ceteris paribus*.